

ADVANCED PEOPLE SENSOR

APS for Retail and Smart Spaces

Reference Manual REST



Original document

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Document History

Version	Modification
2.6.0	Chapter 3 ■ Authorization attempts with incorrect credentials block login for 20 seconds. Chapter 5 ■ New Chapter with to request complete parameter export to JSON ⁺ ■ New Chapter with to request web interface token Chapter 6 ■ New class "Staff" added for the UWB ⁺ staff detection feature. ■ Removed wireless tracking with WiFi and Bluetooth. Chapter 8 ■ New Chapter with curl example Deprecated REST API v1 partly removed, only services for APS Device Manger still included for compatibility reasons with features for update, get diagnostic data and get image snapshots.
2.4.0	Chapter 7.2 ■ Correction: Example for the URL GET with special period Updated several requests
2.2.0	Different chapters ■ New: Implementation new API version 2.0
2.0.0	Chapter 2.1 ■ Change: New permissions and more than 2 user Chapter 4 ■ Change: Old chapter Update now chapter 5 ■ New Chapter: User and rights management Chapter 5 ■ Change: Old chapter Sensor data now chapter 8 Chapter 6 ■ Change: Old chapter Examples now chapter 9 ■ New Chapter: System commands Chapter 7 ■ Change: Old chapter Appendix now chapter 10 ■ New Chapter: System data

Version	Modification
1.18.0	Chapter 5 ■ New classes "Group" and "Cart" for counting with the ObjectClassification-License. ■ New class "Cart" for zone monitoring with the ObjectClassification-License. ■ Wireless tracking for WiFi and Bluetooth with the WirelessServices-License and the USB WiFi Bluetooth Adapter.
1.16.0	New layout
1.15.1	Chapter 3 ■ Removed hint for token update by GET method (still not included today). Chapter 5 ■ New subsection Zone object list - all zones ■ New subsection Zone object list - single zones
1.14.0	Chapter 5 ■ New subsection Zone statistics - all zones ■ New subsection Zone statistics - single zones ■ New subsection System information moved from Chapter 6 Chapter 6 changed to Examples ■ New subsection Integrated python script moved from subsection 2.5 ■ New subsection Tool Insomnia
1.12.0	No change of REST protocol
1.10.1	Correction of testClient.py help
1.10.0	First version of document

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1 INTRODUCTION

Starting with firmware version 2.0, the REST API offers the possibility to retrieve data from the sensors as well as status information. Furthermore, it contains system calls to support relevant maintenance operations.

Representational state transfer (REST) is a stateless software architecture that reads webpages as data. The REST API integrated in HELLA APS⁺ sensors answers to requests from a server. This server needs to have access to the sensors by IP protocols. Therefore, the sensors should not be hidden behind a NAT⁺, for example behind a router in a shop.

The data is encapsulated in JSON⁺ (JavaScript Object Notation).

Since firmware version 2.2 the new extended REST API version 2.0 has been implemented. The old version 1.0 is removed in firmware version 2.6.

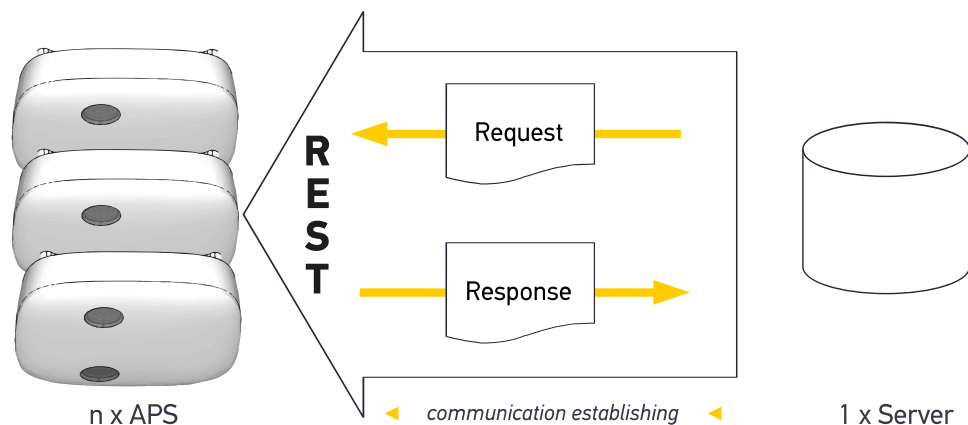


Fig. 1: APS server communication via REST



No Part

The server is not part of the HELLA Aglaia offer.

Support

Service requests can be sent to the following e-mail address:

support@people-sensing.com

You can also use our support center at: ➔ support.people-sensing.com.

The general contact information is shown on page 2.

2 GENERAL INFORMATION

2.1 User and Access Rights

The REST API has the same users as the web interface. With version 2.0 the user and rights management was changed. Depending on the user accessing the interface, only certain data is transmitted or access is denied.

Any number of users with their own password can be set up. Up to 7 different permissions can be assigned to each user:

Group Name in REST	Name in the User Interface	Description
video	Video access	- Can see live images. - Can start/stop video recording via HMI ⁺. - Can download video data via HMI.
countadm	Camera setup	Can set up camera functions (position/angle, counting lines, etc.).
ifadm	Network setup	Can make changes to settings (network, interfaces, updates, import licences, etc).
videoadm	Video setup	Can set up video recording and video streaming.
datauser	Data user	Can view, collect and delete user data
sysadm	System administration	- Can manage users (set up, change, delete, reset passwords) - Can make a reset to Factory Defaults
sshuser	SSH ⁺ access	Can use ssh access, if this service is enabled.

2.2 Data

The data is encapsulated in JSON ⁺ (JavaScript Object Notation).

To identify incoming data, use the sensor's unique MAC ⁺ address or the configurable hostname.

2.3 Data Security

The REST API will only be available for communication via https ⁺.

By default a self-signed certificate is used. For details, see "APS-Customizing.pdf" at chapter "5.5. SSL certificate".

2.4 Versioning

The version is part of the URL ⁺ using `apiV2`. A new version will only be created, if changes are not backward compatible. Adding new requests to the API will not lead to a new version.

2.5 Base Path

The authentication URL of the REST API is:

```
https://[deviceIP]:[restPort]/auth
```

Example

Path to request the bearer token for authentication

```
https://192.168.100.10:8091/auth
```

The base path URL ⁺ of the REST API is:

```
https://[deviceIP]:[restPort]/apiv2/
```

Example

This example is only the path – the call request itself needs bearer token returned from the authentication

```
https://192.168.100.10:8091/apiv2/sensorData
```

2.6 Licenses

There are some optional licenses for the APS, that have effects in the availability of data.

License	Effect	Comments
Object Classification	usage of data	With this license data is separated in the classes: ■ Adult, ■ Child, ■ Group, ■ Cart, ■ Staff. Without this license the data at <code>Adult</code> contains the sum of <code>Adult</code> and <code>Child</code>. The <code>Adult</code> object class is the only one output.
Zone Monitoring	required for data types	■ <code>zoneObjectList</code>, ■ <code>zoneEvent</code>, ■ <code>zoneAlert</code>.
Object List	required for data type	■ <code>objectList</code>.

3 AUTHENTICATION

Authorization is performed via JWT tokens. Username and password are required to receive the initial token. A token is valid for 1 hour.

Two failed attempts with incorrect credentials are tolerated. After the third failed attempt, the login will be blocked for 20 seconds between each new incorrect attempt to avoid brute force attacks.

3.1 Token Information

Part	Content	Comment
HEADER	"typ": "JWT", "alg": "HS256"	Token type → JsonWebToken Signature type → HMAC-SHA256
PAYLOAD	"iat": 1492765233, "exp": 1492768833, "nbf": 1492765233, "identity": 2	Time of creation Valid until Valid from User (2 → user-role-edit)
VERIFY SIGNATURE	HMACSHA256(base64UrlEncode(header) + "." + base64UrlEncode(payload) , secretKey)	

3.2 Auth

Name	auth	
URL +	https://[deviceIP]:[restPort]/auth	
Availability	APS-RS firmware 1.10	
Permission	permission required since APS-RS firmware 2.0	
	POST	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	—	not supported	405 - Method Not Allowed	
POST	User / Password	Token in JWT format	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 429 - Too Many Requests in case of too many invalid attempts and block login for 20 seconds	
PUT	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

3.3 Example for Initial Token Request

Request

Example

```
POST /auth HTTP/1.1
Host: 192.168.100.10
Connection: keep-alive
Content-Length: 48
Origin: http://192.168.100.10
Content-Type: application/json; charset=UTF-8
Referer: http://192.168.100.10/
{
  "username": "admin",
  "password": "admin"
}
```

Response

Example

```

HTTP/1.1 200 OK
< Server: TornadoServer/6.0.4
< Content-Type: application/json
< Date: Mon, 16 Jul 2023 12:04:42 GMT
< Etag: "bbcc174ac65c1260527cfa01d91f151965cd6537"
< Content-Length: 244

{
  "access_token":
  "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpZGVudGl0eSI6MCwiaWF0IjoxNjA1NTI3NTAwLCJleHAiOiJlMDU1MzExMDAsIm5iZiI6MTYwNTUyNzUwMH0.BgiERjtIqXDU0PHNlOAd0mFG8TNOEeF27-hn-8iGc6I"
}
```

3.4 Using the JSON Web Token

All requests except for auth/POST require the keyword Bearer and the token in the Authorization field of the header.

Example for a Request Using a Token

```

GET /apiv2/sensorData/counts HTTP/1.1
Host: 192.168.100.10
Connection: keep-alive
Authorization: Bearer
eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJleHAiOiJlMDU1MzExMDAsIm5iZiI6MTQ5Mjc3ODY4OCwiaWRlbnRpdHkiOiJlMDU1MzExMDY4OH0.aYEc3YNgkGDYEIXXVb5BPKyDTWrRzx100zxOuqOngco
Referer: http://192.168.100.10/
```

4 USER AND RIGHTS MANAGEMENT

With the user and rights management, users can be queried, created and edited.

4.1 All User

Request

Name	users	
URL ⁺	https://[deviceIP]:[restPort]/apiv2/users	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	sysadm Each user sees his own data
	POST	sysadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with the data of all users for a sysadmin or the user information of the own user fo all other per-missions	200 - OK on success 400 - Bad Request on internal error → additional informa-tion in JSON error payload 401 - Unauthorized — The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	JSON Container with data of the new user	—	200 - OK on success 400 - Bad Request on internal error → additional informa-tion in JSON error payload 401 - Unauthorized — The request requires an user authentication 403 - Permission denied — The user requires sysadm permission	
DELETE	—	not supported	405 - Method Not Allowed	

Response

GET returns the following JSON container with the data of all known users

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timestamp": "2023-11-01T16:42:00+02:00",
  "users": {
    "1000": {
      "username": "admin",
      "id": 1000,
      "gid": 65534,
      "groups": [
        1001,
        2000,
        2001,
        2002,
        2004,
        2005,
        2006
      ],
      "groupNames": [
        "video",
        "countadm",
        "ifadm",
        "sysadm",
        "datauser",
        "videoadm",
        "sshuser"
      ],
      "isLocked": false,
      "expires": -1,
      "lastChange": 18857,
      "password": "",
      "passwordExpiresImmediately": false
    },
    "1001": {
      "username": "network_setup",
      "id": 1001,
      "gid": 65534,
      "groups": [
        2001,
        2006
      ],
      "groupNames": [
        "ifadm",
        "sshuser"
      ],
      "isLocked": false,
```

```

    "expires": -1,
    "lastChange": 18859,
    "password": "",
    "passwordExpiresImmediately": false
  },
  "1002": {
    "username": "image_setup",
    "id": 1002,
    "gid": 65534,
    "groups": [
      1001,
      2000,
      2004
    ],
    "groupNames": [
      "video",
      "countadm",
      "datauser"
    ],
    "isLocked": false,
    "expires": -1,
    "lastChange": 18859,
    "password": "",
    "passwordExpiresImmediately": false
  }
}

```

Name	Type	Description
username	string	Username
id	integer	User ID
gid	integer	Primary Group ID, can be ignored.
groups	array of integer	Field with user group IDs.
group_names	array of string	Field with names of the user's groups.
isLocked	boolean	Password Lock (lock for corresponding user can be set and reset by sysadmin).
expires	integer	Specifies a time in days after 1 jan 1970 in which a login is possible. If the value is -1, the login does not expire.
lastChange	integer	When was the password last changed (in days since 1.1.1970). If the value is 0 then password must be changed at next login. The evaluation of the 0 should not be ignored in the REST API when authenticating.
password	string	In that case always empty.
passwordExpiresImmediately	boolean	Setting whether the password must be changed at the next login.

Content of a POST Request

A POST on the route creates a new user.

Name	Type	Default	Mandatory	Description
username	string	nothing	Yes	Username
groupNames	array of string	[]	No	Field with names of the user's groups
isLocked	boolean	false	No	Password Lock (lock for corresponding user can be set and reset by sysadmin)
password	string	nothing	Yes	Password
passwordExpiresImmediately	boolean	false	No	Setting whether the password must be changed at the next login.

Rules for username

- A user name has 1-32 characters.
- A user name starts with a lowercase "a-z" or an underscore "_".
- A user name consists only of lowercase letters "a-z", numbers "0-9", underscores "_", minuses "-" and dollar signs "\$".
- Only the last character in the user name can be a dollar sign "\$".

Group Name in REST	Name in the User Interface	Description
video	Video access	■ Can see live images. ■ Can start/stop video recording via HMI +. ■ Can download video data via HMI.
countadm	Camera setup	Can set up camera functions (position/angle, counting lines, etc.).
ifadm	Network setup	Can make changes to settings (network, interfaces, updates, import licences, etc.).
videoadm	Video setup	Can set up video recording and video streaming.
datauser	Data user	Can view, collect and delete user data
sysadm	System administration	■ Can manage users (set up, change, delete, reset passwords) ■ Can make a reset to Factory Defaults
sshuser	SSH access	Can use ssh access, if this service is enabled.

Example

User With 3 Groups

```
{
  "username": "myuser1",
  "groupNames": ["video", "ifadm", "sysadm"],
  "isLocked": false,
  "expires": "-1",
  "password": "x",
  "passwordExpiresImmediately": true
}
```

4.2 Specific User

Request

Name	users/SpecificUser	
URL +	https://[deviceIP]:[restPort]/apiv2/users/myuser1	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	sysadm Each user sees his own data.
	PUT	sysadm Every user is allowed to change his own Password.
	DELETE	sysadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON Container with data of the user	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 403 - Permission denied — The user requires sysadm permission 404 - Not found — User not available	
PUT	JSON Container with data of the specific user as sysadm or with new password for all other permissions	NONE	200 - OK on success 403 - Permission denied — The user requires sysadm permission 404 - Not found — User not available 405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	NONE	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 403 - Permission denied — The user requires sysadm permission 405 - Method Not Allowed	Delete the specific user, if the permission exists

Response

GET returns the following JSON container with the data of the specific user

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timestamp": "2023-11-01T16:42:00+02:00",
  "userInfo": {
    "username": "myuser1",
    "id": 1007,
    "gid": 65534,
    "groups": [
      1001,
      2001,
      2002
    ],
    "groupNames": [
      "video",
      "ifadm",
      "sysadm"
    ],
    "isLocked": false,
    "expires": -1,
    "lastChange": 0,
    "password": "",
    "passwordExpiresImmediately": true
  }
}
```

Contents of a PUT Request

A PUT on the route changes the following Settings of the specific user:

- Password.
- Group membership.

Name	Type	Default	Mandatory	Description
group_names	array of string	[]	No	Field with names of the user's groups
isLocked	boolean	false	No	Password Lock (lock for corresponding user can be set and reset by sysadmin)
password	string	nothing	No	Password, if the password is empty, the old password will be kept.
passwordExpire- immediately	boolean	false	No	Setting whether the password must be changed at the next login.

Using group_names all memberships be deleted and the new memberships will be set

Group Name in REST	Name in the User Interface	Description
video	Video access	■ Can see live images. ■ Can start/stop video recording via HMI ⁺. ■ Can download video data via HMI.
countadm	Camera setup	Can set up camera functions (position/angle, counting lines, etc.).
ifadm	Network setup	Can make changes to settings (network, interfaces, updates, import licences, etc).
videoadm	Video setup	Can set up video recording and video streaming.
datauser	Data user	Can view, collect and delete user data
sysadm	System administration	■ Can manage users (set up, change, delete, reset passwords) ■ Can make a reset to Factory Defaults
sshuser	SSH access	Can use ssh access, if this service is enabled.

Content of a DELETE Request

A DELETE on the route deletes the specified user.

- To delete a user a sysadm permission is necessary.
- It is not possible to delete the own user.

5 SYSTEM COMMAND

5.1 Update

To perform a firmware update post the URL ⁺ to the update .tar file to the device.

Request

Name	update	
URL ⁺	https://[deviceIP]:[restPort]/apiv2/systemCommand/update	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	PUT	ifadm

Method	Data	Result	HTTP Status	Comments
GET	—	not supported	405 - Method Not Allowed	
PUT	JSON Container with update information	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Contents of a PUT Request

- updateType
 - Enumeration specifying the type of update
 - Currently only "UPDATE_FIRMWARE" is supported
- packetUrl
 - String, maxLen: 255 characters
 - URL to the update file
- Validity
 - Unsigned integer
 - Min: 1h, max: 168 h (one week), default: 48h
 - Specifies the time period, in which the device shall perform retries in case an update fails

Example

Example Update

```
{
  "requestData":
  {
    "updateType": "UPDATE_FIRMWARE",
    "packetUrl": "http://firmwareUpdateSite.org/firmwareUpdate.tar",
    "validity": 1
  }
}
```

5.2 Diagnostic Data

Request

Name	diagnosticData	
URL +	https://[deviceIP]:[restPort]/apiv2/systemCommand/diagnosticData	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm Only with video permission the image snapshots are included in the diagnostic data.

Method	Data	Result	HTTP Status	Comments
GET	NONE	Binary file: Package with diagnostic data as .tar archive	200 - OK on success 400 - Bad Request on internal error → additional informa- tion in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

5.3 Image Snapshots

Request

Name	imageSnapshots	
URL +	https://[deviceIP]:[restPort]/apiv2/systemCommand/imageSnapshots	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video

Method	Data	Result	HTTP Status	Comments
GET	NONE	Binary file: Package with image data as .tar archive	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload ■ imagegrabbing error - error while grabbing the images ■ tar error - error while creating the .tar archive ■ unknown/other error - unknown or other error 401 - Unauthorized – The request requires an user authentication	The .tar archive contains the following image files: ■ Original left image ■ Right rectified image ■ Disparity map
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

5.4 Parameter Set

Request

Name	parameterSet
URL +	https://[deviceIP]:[restPort]/apiv2/systemCommand/parameterSet
Availability	APS-RS firmware 2.2
Permission	permission required since APS-RS firmware 2.0
GET	videoadm, countadm, ifadm, sysadm
PUT	videoadm, countadm, ifadm, sysadm
	Only the parameter files for which the permission is sufficient are overwritten.

Method	Data	Result	HTTP Status	Comments
GET	NONE	Binary file: Package with exported parameter files as .tar archive	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	The archive and the data in it are created at the time when the request was made.
PUT	Binary file: Package with parameter files to be imported as .tar archive. And JSON container with parameter import details.	JSON container with exit code and the outputs of the script on stdout as well as stderr.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	Important! The import of the parameter set shows effect only after a reboot!
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

A GET on the route returns an archive with the current parameter files of the device.

A PUT on the route imports an archive of parameter files into the device and replaces the previous configuration.

5.5 Parameter Export

Request

Name	configuration	
URL +	https://[deviceIP]:[restPort]/apiv2/systemCommand/configuration	
Availability	APS-RS firmware 2.6	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	—	JSON container with with the complete set of parameters	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET request on the route returns a JSON container with the complete set of parameters.

5.6 Setup Video Recording

Request

Name	setUpVideoRecording	
URL +	https://[deviceIP]:[restPort]/apiv2/systemCommand/setupVideoRecording	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	videoadm
	PUT	videoadm
	DELETE	videoadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with current recording setup	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	If no timer is set the container is empty
PUT	JSON container with new recording setup	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
POST	—	not supported	405 - Method Not Allowed	
DELETE	NONE	NONE	204 - OK, The resource was successfully deleted on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	

A DELETE on the route resets any existing recording settings.

Response

A GET request on the route returns a JSON container with any recording settings that may be present.

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timestamp": "2023-11-01T16:42:00+02:00",
  "setUpVideoRecording": {
    "timer": {
      "month": 5,
      "day": 7,
      "hour": 13,
      "minute": 32
    },
    "recDuration": 60
  }
}
```

If no timer is set, the JSON object is empty.

Example for GET With No Timer

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timestamp": "2023-11-01T16:42:00+02:00",
  "setUpVideoRecording": {}
}
```

Contents of a PUT Request

A PUT on the route sets a new recording setting located in the JSON (possibly overwriting existing ones).

Example for Setting New Exposure Settings

```
{
  "timer": {
    "month": 6,
    "day": 7,
    "hour": 13,
    "minute": 32
  },
  "rec_duration": 60
}
```

5.7 Reboot

Request

Name	reboot	
URL +	https://[deviceIP]:[restPort]/apiv2/systemCommand/reboot	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	PUT	videoadm, countadm, ifadm, sysadm

Method	Data	Result	HTTP Status	Comments
GET	—	not supported	405 - Method Not Allowed	
PUT	NONE	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 501 - Not implemented	Device reboots
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

A PUT on the route will cause the device to reboot in the following seconds. Receipt of the message is acknowledged by means of (200 - OK). The restart takes place afterwards.

Restarting the device leads to a temporary disconnection with the device. In this case, the client software must establish a new session and be implemented accordingly.

5.8 Power Cycle

Request

Name	powerCycle	
URL +	https://[deviceIP]:[restPort]/apiv2/systemCommand/powerCycle	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	PUT	videoadm, countadm, ifadm, sysadm

Method	Data	Result	HTTP Status	Comments
GET	—	not supported	405 - Method Not Allowed	
PUT	NONE	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 501 - Not implemented – Power cycle with this device not possible	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

5.9 Find Me

Request

Name	findMe	
URL +	https://[deviceIP]:[restPort]/apiv2/systemCommand/findMe	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	videoadm, countadm, ifadm, sysadm
	PUT	videoadm, countadm, ifadm, sysadm

Method	Data	Result	HTTP Status	Comments
GET	—	JSON container with actual FindMe status	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	JSON Container specifying whether the LED flashing should be started or stopped.	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET request on the route returns a JSON container with current FindMe status ("state":"started"|"stopped"|"unknown").

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timeStamp": "2023-11-01T16:42:00+02:00",
  "findMe": {
    "state": "started"
  }
}
```

Contents of a PUT Request

A PUT on the route starts or stops the flashing of the LED on the device

Example for Start Flashing

```
{
  "command": "start"
}
```

Example for Stop Flashing

```
{
  "command": "stop"
}
```

5.10 Web Interface Token

Request

Name	webInterfaceToken	
URL ⁺	https://[deviceIP]:[restPort]/apiv2/systemCommand/webInterfaceToken	
Availability	APS-RS firmware 2.6	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	—	JSON container with token	200 - OK on success 400 - Bad Request on internal error → additional information only inside the sensor in RestApi.log and syslog 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET request on the route returns a JSON container with a token. Using this token, the user currently logged in to the REST API can open the web interface / HMI of the sensor with the same credentials.

This token

- can only be used once,
- can only be used on the device on which it was created and
- only valid for a limited time (1 minute).

Example for GET

```
{
  "token":
  "eyJhbGciOiJIUzUxMiIsInR5cCI6IkpXVCJ9.eyJtYWMiOiIwMDowQjo5MTTo5MDo5Mjo2RiIsImV4cCI6MTY5MjAyMTc0NywiaWF0IjoxNjkyMDIxNjg3LCJuYmYiOiJlY2OTIwMjE2ODd9.Gc0JGE5enla_w3_aUcbUXKDCNpyEgNh19KtrajQqsq1dZ73FnAlGsngxuSDDyxUokLxkzr9IemxDcTarwzmA_mg"
}
```

Use the IP address of the sensor and this token to open the HMI of the sensor:

Example as URL to the HMI

```
http://192.168.100.65/?
token=eyJhbGciOiJIUzUxMiIsInR5cCI6IkpXVCJ9.eyJtYWMiOiIwMDowQjo5MTTo5MDo5Mjo2RiIsImV4cCI6MTY5MjAyMTc0NywiaWF0IjoxNjkyMDIxNjg3LCJuYmYiOiJlY2OTIwMjE2ODd9.Gc0JGE5enla_w3_aUcbUXKDCNpyEgNh19KtrajQqsq1dZ73FnAlGsngxuSDDyxUokLxkzr9IemxDcTarwzmA_mg"
}
```

6 SYSTEM DATA

6.1 All System Data

Request

Name	systemData	
URL +	https://[deviceIP]:[restPort]/apiv2/systemData	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with all system data.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET returns a JSON container with all system data:

Example for GET

```
{
  "systemData": {
    "timestamp": "2023-11-01T16:42:00+02:00",
    "hostname": "APS901234",
    "licenses": [
      {
        "feature": "ObjectClassification",
        "valid": true
      },
      {
        "feature": "ObjList",
        "valid": true
      },
      {
        "feature": "VideoRecording",
```

```
        "valid": true
    },
    {
        "feature": "ZoneMonitoring",
        "valid": true
    },
    {
        "feature": "WirelessServices",
        "valid": true
    }
],
"firmwareVersion": "2.6.0.0",
"modelType": "APS-90-IO-8GB",
"networkAdapters": [
    {
        "name": "eth0",
        "macAddress": "00:0B:91:90:12:34",
        "inetAddress": "192.168.100.65",
        "mask": "255.255.255.0"
    },
    {
        "name": "wlan0",
        "macAddress": "FE:88:99:77:66:12",
        "inetAddress": "172.88.100.23",
        "mask": "255.255.255.0"
    },
    {
        "name": "tun0",
        "macAddress": "00:00:00:00:00:00",
        "inetAddress": "10.8.0.123",
        "mask": "255.255.255.0"
    }
],
"errorState": {
    "osdState": "Ok"
},
"systemState": {
    "currentUptime": {
        "days": 0,
        "hours": 0,
        "minutes": 3109
    },
    "operationHoursTotal": 10055,
    "operationHoursUpdate": 44,
    "reasonForReboot": "normal reset (by an application)",
    "totalReboots": 2009,
    "powerCycles": 128,
    "watchdogEventsHW": 0,
    "watchdogEventsSW": 1,
    "cpuTemperature": 58
}
```

```

    }
  }
}

```

6.2 Time

Request

Name	time	
URL +	https://[deviceIP]:[restPort]/apiv2/systemData/time	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm
	PUT	ifadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with current system date and system time.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	JSON container with new system date and system time.	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET returns the current system time:

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timestamp": "2023-11-01T16:42:00+02:00",
}
```

Contents of a PUT Request

A PUT sets the system time:

Example to Set the System Time

```
{
  "timestamp": "2023-11-20T13:12:31+00:00"
}
```

The string for the time specification must be ISO 8601 compliant and may contain time zone information. In this case, the offset is offset to UTC time with the time zone of the sensor.



The time zone of the device cannot be changed via the REST API.

Interpretation of the time specification on the device:

- Time without time zone information: "time": "2023-11-20T13:24:56".
 - The time is interpreted and set as localtime.
- Time with time zone information: "time": "2023-11-20T13:24:56+02:00".
 - The specified time offset to UTC is calculated according to the time zone of the sensor.
- Specified as UTC: "time": "2023-11-20T13:24:56Z".
 - The specified time is calculated according to the time zone of the sensor.

6.3 Host Name

Request

Name	hostname	
URL +	https://[deviceIP]:[restPort]/apiv2/systemData/hostname	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm
	PUT	ifadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with current host name.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	JSON container with new host name.	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET on the route returns a JSON with the current hostname.

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timeStamp": "2023-11-01T16:42:00+02:00",
}
```

Contents of a PUT Request

A PUT on the route sets a new hostname.

Example to Set the Host Name

```
{
  "hostname": "MyAPS"
}
```

6.4 Licenses

Request

Name	licenses	
URL +	https://[deviceIP]:[restPort]/apiv2/systemData/licenses	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with information about availability and status of all licenses on the device.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET on the route returns a JSON with all licenses available on the device

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timeStamp": "2023-11-01T16:42:00+02:00",
  "licenses": [
    {
      "feature": "ObjectClassification",
      "valid": true
    },
    {
      "feature": "ObjectList",
      "valid": true
    },
    {
      "feature": "VideoRecording",
      "valid": true
    }
  ]
}
```

If no licenses are installed, the licenses block remains empty.

Example for no licences

```
{
  "macAddress": "00:0B:91:90:91:92",
  "hostname": "APS909192",
  "timeStamp": "2023-11-01T16:42:00+02:00",
  "licenses": []
}
```


6.5 Firmware Version

Request

Name	firmwareVersion	
URL +	https://[deviceIP]:[restPort]/apiv2/systemData/firmwareVersion	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON Container with the firmware version of the device	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET on the route returns a JSON with the firmware version string.

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timeStamp": "2023-11-01T16:42:00+02:00",
  "firmwareVersion": "2.6.0.0"
}
```

6.6 Model Type

Request

Name	modelType	
URL +	https://[deviceIP]:[restPort]/apiv2/systemData/modelType	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON Container with data of all users	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET on the route returns a JSON with the model name.

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timeStamp": "2023-11-01T16:42:00+02:00",
  "modelType": "APS-90E-IO"
}
```

6.7 Network Adapters

Request

Name	networkAdapters	
URL +	https://[deviceIP]:[restPort]/apiv2/systemData/networkAdapters	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON Container with MAC addresses	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET on the route returns a JSON with the following information about the existing network adapters:

- Name
- MAC address
- IP
- Subnetmask

Unavailable network adapters (adapters without IP) are not displayed.

The loopback device is not displayed.

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timestamp": "2023-11-01T16:42:00+02:00",
  "networkAdapters": [
    {
      "name": "eth0",
      "macAddress": "00:0B:91:90:12:34",
      "inetAddress": "192.168.100.65",
      "mask": "255.255.255.0"
    },
    {
      "name": "wlan0",
      "macAddress": "FE:88:99:77:66:12",
      "inetAddress": "172.88.100.23",
      "mask": "255.255.255.0"
    },
    {
      "name": "tun0",
      "macAddress": "00:00:00:00:00:00",
      "inetAddress": "10.8.0.123",
      "mask": "255.255.255.0"
    }
  ]
}
```

6.8 Error State

Request

Name	errorState	
URL +	https://[deviceIP]:[restPort]/apiv2/systemData/errorState	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON Container with error states	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET on the route returns a JSON with the current error states currently OSD only.

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timestamp": "2023-11-01T16:42:00+02:00",
  "errorState": {
    "osdState": "OSD_STATE_ERROR_COVERED"
  }
}
```

Result codes of the Optical self-diagnosis (OSD) are listed in ➔ Chapter 9.1 OSD Status Codes on page 82.

6.9 System State

Request

Name	systemState	
URL +	https://[deviceIP]:[restPort]/apiv2/systemData/systemState	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	video, videoadm, countadm, ifadm, datauser, sysadm

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON Container with system states	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

A GET on the route returns a JSON with the current system states

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "timeStamp": "2023-11-01T16:42:00+02:00",
  "systemState": {
    "currentUptime": {
      "days": 12,
      "hours": 6,
      "minutes": 17
    },
    "operationHoursTotal": 10055,
    "operationHoursUpdate": 44,
    "reasonForReboot": "normal reset (by an application)",
    "totalReboots": 2010,
    "powerCycles": 128,
    "watchdogEventsHW": 0,
    "watchdogEventsSW": 1,
  }
}
```

```
}  
  "cpuTemperature": 58  
}
```

7 SENSOR DATA

7.1 Sensor Data - All Available Data

On this request, the sensor returns all available data. Data is only transmitted if the corresponding licenses (Object Classification license, Zone Monitoring license, Wireless Services license, etc.) are available and configurations (cart counting, group counting, etc.) are active.

Request

Name	sensorData	
URL ⁺	https://[deviceIP]:[restPort]/apiv2/sensorData	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	datauser

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with sensor data	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	Returns all available sensor data
POST	—	not supported	405 - Method Not Allowed	
PUT	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

Example for GET

```
{
  "sensorData": {
    "macAddress": "00:0B:91:90:12:34",
    "hostname": "APS901234",
    "countingData": {
      "timeStamp": "2023-11-22T16:42:00+02:00",
      "countingLines": [
        {
          "name": "MyLine0",
          "data": [
            {
              "class": "Adult",
              "in": 0,
              "out": 0
            },
            {
              "class": "Child",
              "in": 0,
              "out": 0
            }
          ]
        },
        {
          "name": "MyLine1",
          "data": [
            {
              "class": "Adult",
              "in": 0,
              "out": 0
            },
            {
              "class": "Child",
              "in": 0,
              "out": 0
            }
          ]
        }
      ]
    },
    "fillLevel": {
      "timeStamp": "2023-11-22T16:42:00+02:00",
      "zones": [
        {
          "name": "MyZone0",
          "data": [
            {
              "class": "Adult",
              "level": 0,
              "avgDwell": 0,
              "maxDwell": 0
            },
            {
              "class": "Child",
              "level": 0,
```

```

        "avgDwell": 0,
        "maxDwell": 0
    } ]
},
{
    "name": "MyZone1",
    "data": [
        {
            "class": "Adult",
            "level": 0,
            "avgDwell": 0,
            "maxDwell": 0
        },
        {
            "class": "Child",
            "level": 0,
            "avgDwell": 0,
            "maxDwell": 0
        }
    ]
} ]
},
"zoneStatistics":
{
    "timeStamp": "2023-11-22T16:42:00+02:00",
    "interval": 300,
    "zones": [
        {
            "name": "MyZone0",
            "data": [
                {
                    "class": "Adult",
                    "count": 0,
                    "maxDwell": 0,
                    "maxFillLevel": 0,
                    "meanDwell": 0
                },
                {
                    "class": "Child",
                    "count": 0,
                    "maxDwell": 0,
                    "maxFillLevel": 0,
                    "meanDwell": 0
                }
            ]
        },
        {
            "name": "MyZone1"
            "data": [
                {
                    "class": "Adult",
                    "count": 0,
                    "maxDwell": 0,
                    "maxFillLevel": 0,
                    "meanDwell": 0
                },
                {
                    "class": "Child",
                    "count": 0,

```

```

        "maxDwell": 0,
        "maxFillLevel": 0,
        "meanDwell": 0
    } ]
} ]
},
"zoneObjectList":
{
    "timeStamp": "2023-11-22T16:48:00+02:00",
    "interval": 300,
    "zones": [
    {
        "name": "BlueZone",
        "objects": [
        {
            "id": 2,
            "class": "Adult",
            "dwell": 15,
            "exitTime": "2023-11-20T12:04:36.025271+01:00",
            "height": 176,
            "entry": {
                "x": 103,
                "y": 10
            },
            "exit": {
                "x": 658,
                "y": 328
            }
        },
        {
            "id": 1,
            "class": "Child",
            "dwell": 345,
            "exitTime": "2023-11-20T12:02:36.025271+01:00",
            "height": 163,
            "entry": {
                "x": 238,
                "y": 600
            },
            "exit": {
                "x": 658,
                "y": 680
            }
        }
        ]
    },
    {
        "name": "RedZone",
        "objects": []
    }
    ]
},
"wirelessLogging":

```

```
{
  "timeStamp": "2023-11-03T13:32:13+00:00",
  "stations": [
    {
      "MAC": "D8:E0:E1:33:C7:EF",
      "Type": "BLUETOOTH",
      "Signal": -76,
      "Time": "2023-11-02T11:30:06.075467+00:00"
    },
    {
      "MAC": "FF:FF:C2:07:BA:66",
      "Type": "BLUETOOTH",
      "Signal": -58,
      "Time": "2023-11-02T11:30:06.929436+00:00"
    }
  ]
}
```

7.2 Count Results - All Counting Lines

Request or reset the counting data of all counting lines.

Request

Name	countingData	
URL +	https://[deviceIP]:[restPort]/apiv2/sensorData/countingData	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	datauser
	PUT	datauser

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with count results	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	Returns accumulated count results of all counting lines
GET	Two time stamps as parameters in the URL	JSON container with the count data of the requested interval.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	
PUT	NONE	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication	All count results will be set to zero (but only for the Rest interface)
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

Example for GET Without Data in URL

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "countingData":
  {
    "timeStamp": "2023-11-01T16:42:00+02:00",
    "countingLines": [
      {
        "name": "MyLine0",
        "data": [
          {
            "class": "Adult",
            "in": 0,
            "out": 0
          }
        ],
      }
    ],
  }
}
```

```

        {
            "class": "Child",
            "in": 0,
            "out": 0
        } ]
    },
    {
        "name": "MyLine1",
        "data": [
            {
                "class": "Adult",
                "in": 0,
                "out": 0
            },
            {
                "class": "Child",
                "in": 0,
                "out": 0
            } ]
        } ]
    }
}

```

A request for count data for a special period is possible by passing the following parameters with the GET request.

from="2023-11-11T16:00:00+02:00" to="2023-11-11T17:00:00+02:00"

Example for the URL: <https://192.168.178.61:8091/apiv2/sensorData/countingData?from=2023-11-11T16%3A00%3A00%2B02%3A00&to=2023-11-11T17%3A00%3A00%2B02%3A00>

Example for GET With Special Period

```

{
    "macAddress": "00:0B:91:90:91:92",
    "hostname": "APS909192",
    "countingData":
    {
        "timeStamp": "2023-11-11T17:00:00+02:00",
        "interval": 3600,
        "countingLines": [
            {
                "name": "MyLine0",
                "data": [
                    {
                        "class": "Adult",
                        "in": 0,
                        "out": 0
                    },
                    {
                        "class": "Child",
                        "in": 0,

```

```
        "out": 0
      } ]
    },
    {
      "name": "MyLine1",
      "data": [
        {
          "class": "Adult",
          "in": 0,
          "out": 0
        },
        {
          "class": "Child",
          "in": 0,
          "out": 0
        } ]
      } ]
    }
  }
```

7.3 Count Results - Single Counting Line

Request or reset the counting data of a single counting line.

Request

Name	countingData/MyLine0	
URL +	https://[deviceIP]:[restPort]/apiv2/sensorData/countingData/[countingLineName]	
Availability	APS-RS firmware 2.2	
Permission	permission required since APS-RS firmware 2.0	
	GET	datauser
	PUT	datauser

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with count results	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized — The request requires an user authentication 404 - Not found — There is no resource behind the URI→When counting line is not present	Returns accumulated count results of specific counting line
GET	Two time stamps as parameters in the URL	JSON container with the count data of the requested interval.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found — There is no resource behind the URI→When counting line is not present	
PUT	NONE	NONE	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found — There is no resource behind the URI	Count results of specific counting line will be set to zero (but only for the Rest interface)
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

GET without further data returns the following JSON container with the counter values since last boot time or last reset of the counter line.

Example for GET Without Data in URL

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "countingData":
  {
    "timeStamp": "2023-11-11T16:42:00+02:00",
    "countingLines": [
      {
        "name": "MyLine0",
        "data": [
          {
            "class": "Adult",
            "in": 0,
            "out": 0
          },
          {
            "class": "Child",
            "in": 0,
            "out": 0
          }
        ]
      }
    ]
  }
}
```

A request for count data for a special period is possible by passing the following parameters with the GET request.

from="2023-11-11T16:00:00+02:00" to="2023-11-11T17:00:00+02:00"

Example for Get With Period Data in URL

```
{
  "macAddress": "00:0B:91:90:91:92",
  "hostname": "APS909192",
  "countingData":
  {
    "timeStamp": "2023-11-11T17:00:00+02:00",
    "interval": 3600,
    "countingLines": [
      {
        "name": "MyLine0",
        "data": [
          {
            "class": "Adult",
            "in": 0,
            "out": 0
          }
        ]
      }
    ]
  }
}
```

```
    },  
    {  
      "class": "Child",  
      "in": 0,  
      "out": 0  
    } ]  
  } ]  
}
```

If no counting lines are set up, the countingLines block remains empty.

Example for Get Without Configured Counting Lines

```
{  
  "macAddress": "00:0B:91:90:91:92",  
  "hostname": "APS909192",  
  "countingData":  
  {  
    "timeStamp": "2023-11-11T16:42:00+02:00",  
    "interval": 300,  
    "countingLines": []  
  }  
}
```

7.4 Fill Level - All Zones

Request the fill level of all zones.

Request

Name	fillLevel	
URL +	https://[deviceIP]:[restPort]/apiv2/sensorData/fillLevel	
Availability	APS-RS firmware 2.2	
License	Zone Monitoring license required.	
Permission	permission required since APS-RS firmware 2.0	
	GET	datauser

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with zone information	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found — no resource behind the URL → License zone monitoring missing	Returns current fill level of all zones.
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "fillLevel":
  {
    "timestamp": "2023-11-11T16:42:00+02:00",
    "zones": [
      {
        "name": "MyZone0",
        "data": [
          {
            "class": "Adult",
            "level": 0,
            "avgDwell": 0,
            "maxDwell": 0
          },
          {
            "class": "Child",
            "level": 0,
            "avgDwell": 0,
            "maxDwell": 0
          }
        ]
      },
      {
        "name": "MyZone1",
        "data": [
          {
            "class": "Adult",
            "level": 0,
            "avgDwell": 0,
            "maxDwell": 0
          },
          {
            "class": "Child",
            "level": 0,
            "avgDwell": 0,
            "maxDwell": 0
          }
        ]
      }
    ]
  }
}
```

7.5 Fill Level - Single Zone

Request the fill level of a specific zones.

Request

Name	fillLevel/myZone0	
URL +	https://[deviceIP]:[restPort]/apiv2/sensorData/fillLevel/[zoneName]	
Availability	APS-RS firmware 2.2	
License	Zone Monitoring license required.	
Permission	permission required since APS-RS firmware 2.0	
	GET	datauser

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with zone information	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - not found — no resource behind the URI → Zone not existing or license zone monitoring missing	Returns current fill level of a specific zone.
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "fillLevel":
  {
    "timeStamp": "2023-11-11T16:42:00+02:00",
    "zones": [
      {
        "name": "MyZone0",
        "data": [
          {
            "class": "Adult",
            "level": 0,
            "avgDwell": 0,
            "maxDwell": 0
          },
          {
            "class": "Child",
            "level": 0,
            "avgDwell": 0,
            "maxDwell": 0
          }
        ]
      }
    ]
  }
}
```

If no zone is set up, the zones block remains empty.

Example for Get Without Configured Zones

```
{
  "macAddress": "00:0B:91:90:91:92",
  "hostname": "APS909192",
  "fillLevel":
  {
    "timeStamp": "2023-11-11T16:42:00+02:00",
    "zones": []
  }
}
```

7.6 Zone Statistics - All Zones

Request zone statistics of all zones.

Request

Name	zoneStatistics	
URL ⁺	https://[deviceIP]:[restPort]/apiv2/sensorData/zoneStatistics	
Availability	APS-RS firmware 2.2	
License	Zone Monitoring license required.	
Publish interval	■ Zone statistics are calculated over an interval of 1, 5, 10, 15 (default), 30, 60 minutes. Interval is configurable via the web interface.	
Permission	permission required since APS-RS firmware 2.0	
	GET	datauser

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with zone statistics	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found - no resource behind the URL → License zone monitoring missing	Returns current zone statistics of all zones.
GET	Two time stamps as parameters in the URL	JSON container with the zone statistics of the requested interval.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found - no resource behind the URL → License zone monitoring missing	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "zoneStatistics":
  {
    "timeStamp": "2023-11-11T16:40:00+02:00",
    "interval": 300,
    "zones": [
      {
        "name": "MyZone0",
        "data": [
          {
            "class": "Adult",
            "count": 0,
            "maxFillLevel": 0,
            "maxDwell": 0
            "meaDwell": 0
          },
          {
            "class": "Child",
            "count": 0,
            "maxFillLevel": 0,
            "maxDwell": 0
            "meanDwell": 0
          }
        ]
      },
      {
        "name": "MyZone1",
        "data": [
          {
            "class": "Adult",
            "count": 0,
            "maxFillLevel": 0,
            "maxDwell": 0
            "meanDwell": 0
          },
          {
            "class": "Child",
            "count": 0,
            "maxFillLevel": 0,
            "maxDwell": 0
            "meanDwell": 0
          }
        ]
      }
    ]
  }
}
```


7.7 Zone Statistics - Single Zone

Request zone statistics of a specific zone.

Request

Name	zoneStatistics/MyZone1	
URL +	https://[deviceIP]:[restPort]/apiv2/sensorData/zoneStatistics/[zoneName]	
Availability	APS-RS firmware 2.2	
License	Zone Monitoring license required.	
Publish interval	■ Zone statistics are calculated over an interval of 1, 5, 10, 15 (default), 30, 60 minutes. Interval is configurable via the web interface.	
Permission	permission required since APS-RS firmware 2.0	
	GET	datauser

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with zone statistic	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found - no resource behind the URL → Zone not existing or license zone monitoring missing	Returns current zone statistics of a specific zone.
GET	Two time stamps as parameters in the URL	JSON container with the zone statistic of the requested interval.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found - no resource behind the URL → Zone not existing or license zone monitoring missing	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "zoneStatistics":
  {
    "timeStamp": "2023-11-11T16:42:00+02:00",
    "interval": 300,
    "zones": [
      {
        "name": "MyZone0",
        "data": [
          {
            "class": "Adult",
            "count": 0,
            "maxFillLevel": 0,
            "maxDwell": 0
            "meaDwell": 0
          },
          {
            "class": "Child",
            "count": 0,
            "maxFillLevel": 0,
            "maxDwell": 0
            "meanDwell": 0
          }
        ]
      }
    ]
  }
}
```

7.8 Zone Object List - All Zones

Request zone object list of all zones.

Request

Name	zoneObjectList	
URL ⁺	https://[deviceIP]:[restPort]/apiv2/sensorData/zoneObjectList	
Availability	APS-RS firmware 2.2	
License	Zone Monitoring license required.	
Publish interval	■ Zone object list are assembled over an interval of 1, 5, 10, 15 (default), 30, 60 minutes. Interval is configurable via the web interface.	
Permission	permission required since APS-RS firmware 2.0	
	GET	datauser

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with the zone object lists of all zones	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found - no resource behind the URL → License zone monitoring missing	Returns current zone object list of all zones.
GET	Two time stamps as parameters in the URL	JSON container with the zone object lists of the requested interval.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found - no resource behind the URL → License zone monitoring missing	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "zoneObjectList":
  {
    "timestamp": "2023-11-11T16:40:00+02:00",
    "interval": 300,
    "zones": [
      {
        "name": "MyZone0",
        "objects": [
          {
            "id": 2,
            "class": "Adult",
            "dwell": 15,
            "exitTime": "2023-11-11T16:36:27+02:00",
            "height": 176,
            "entry": {"x": 103, "y": 10},
            "exit": {"x": 658, "y": 328}
          },
          {
            "id": 1,
            "class": "Child",
            "dwell": 345,
            "exitTime": "2023-11-11T16:36:12+02:00",
            "height": 163,
            "entry": {"x": 238, "y": 600},
            "exit": {"x": 658, "y": 680}
          }
        ]
      },
      {
        "name": "MyZone1",
        "objects": [
          {
            "id": 23,
            "class": "Adult",
            "dwell": 15,
            "exitTime": "2023-11-11T16:38:22+02:00",
            "height": 176,
            "entry": {"x": 103, "y": 10},
            "exit": {"x": 658, "y": 328}
          },
          {
            "id": 17,
            "class": "Child",
            "dwell": 145,
            "exitTime": "2023-11-11T16:39:54+02:00",
            "height": 163,

```

```

        "entry": {"x": 238, "y": 600},
        "exit": {"x": 658, "y": 680}
    }
}

```

In the following example no objects are accumulated in MyZone0 the "objects" field is empty, in MyZone1 2 objects are present.

Example for Get With No Accumulated Objects in MyZone0

```

{
  "macAddress": "00:0B:91:90:91:92",
  "hostname": "APS909192",
  "zoneObjectList":
  {
    "timeStamp": "2023-11-11T16:40:00+02:00",
    "interval": 300,
    "zones": [
      {
        "name": "MyZone0",
        "objects": []
      },
      {
        "name": "MyZone1",
        "objects": [
          {
            "id": 23,
            "class": "Adult",
            "dwell": 15,
            "exitTime": "2023-11-11T16:38:22+02:00",
            "height": 176,
            "entry": {"x": 103, "y": 10},
            "exit": {"x": 658, "y": 328}
          },
          {
            "id": 17,
            "class": "Child",
            "dwell": 145,
            "exitTime": "2023-11-11T16:39:54+02:00",
            "height": 163,
            "entry": {"x": 238, "y": 600},
            "exit": {"x": 658, "y": 680}
          }
        ]
      }
    ]
  }
}

```

If no zone is set up, the zones block remains empty.

Example for Get Without Configured Zones

```
{
  "macAddress": "00:0B:91:90:91:92",
  "hostname": "APS909192",
  "zoneObjectList":
  {
    "timeStamp": "2023-11-11T16:42:00+02:00",
    "interval": 300,
    "zones": []
  }
}
```

7.9 Zone Object List - Single Zone

Request zone object list of a specific zone.

Request

Name	zoneObjectList
URL +	https://[deviceIP]:[restPort]/apiv2/sensorData/zoneObjectList/[zoneName]
Availability	APS-RS firmware 2.2
License	Zone Monitoring license required.
Publish interval	- Zone object list are assembled over an interval of 1, 5, 10, 15 (default), 30, 60 minutes. - Interval is configurable via the web interface.
Permission	permission required since APS-RS firmware 2.0
	GET datauser

Method	Data	Result	HTTP Status	Comments
GET	NONE	JSON container with zone statistics	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found - no resource behind the URL → License zone monitoring missing or zone not existing	Returns current zone object list of a specific zone.
GET	Two time stamps as parameters in the URL	JSON container with the zone object list of the requested interval.	200 - OK on success 400 - Bad Request on internal error → additional information in JSON error payload 401 - Unauthorized – The request requires an user authentication 404 - Not found - no resource behind the URL → License zone monitoring missing or zone not existing	
PUT	—	not supported	405 - Method Not Allowed	
POST	—	not supported	405 - Method Not Allowed	
DELETE	—	not supported	405 - Method Not Allowed	

Response

Example for GET

```
{
  "macAddress": "00:0B:91:90:12:34",
  "hostname": "APS901234",
  "zoneObjectList":
  {
    "timestamp": "2023-11-11T16:42:00+02:00",
    "interval": 300,
    "zones": [
      {
        "name": "MyZone0",
        "objects": [
          {
            "id": 2,
            "class": "Adult",
            "dwell": 15,
            "exitTime": "2023-11-11T16:36:27+02:00",
            "height": 176,
            "entry": {"x": 103, "y": 10},
            "exit": {"x": 658, "y": 328}
          },
          {
            "id": 1,
            "class": "Child",
            "dwell": 345,
            "exitTime": "2023-11-11T16:36:12+02:00",
            "height": 163,
            "entry": {"x": 238, "y": 600},
            "exit": {"x": 658, "y": 680}
          }
        ]
      }
    ]
  }
}
```


8 EXAMPLES

8.1 Integrated Python Script

To see how the REST API works a simple python script is integrated in the device. The APS⁺ uses as standard setup DHCP⁺ - not a fixed IP as his predecessor APC⁺. So first check your e.g. router or OpenDHCP server, which IP the APS uses since startup.

Use telnet or SSH⁺ to navigate inside the sensor and execute the script.

SSH

To use ssh at a Windows PC use tools like e.g. PuTTY (→ <http://www.chiark.greenend.org.uk/~sgtatham/putty/>). At first connecting time you get a "Putty Security Alert", because of a missing ssh key file at your PC - simply accept and store the key by <Yes>. Login with `root` and password `counter`.

Telnet

Telnet is an old and unencrypted protocol. Since firmware 1.6, telnet is by default deactivated. To use it, first activate it in the user interface "Network → Network Services → Telnet".

To connect by telnet at a Windows PC, first activate the telnet feature in "Control Panel", "Programs and Features", "Turn Windows features on or off", and activate the "Telnet Client". Then use the Windows "Command Prompt" and enter:

```
C:\Users\xyz> telnet 192.168.100.10
APS012345 login: root
Password: counter
~ #
```

Use telnet or SSH to navigate inside the sensor and execute the script.

To see how the REST API works, do the following for help:

Example

```
~ # cd /opt/RestApi/test/
/opt/RestApi/test # ./testClient.py -h
usage: testClient.py [-h] [-i IP] [-p PORT] [-U USER] [-P PASSWORD] [-r ROUTE]
                    [-X CMD] [-v]

script for testing REST API routes

optional arguments:
  -h, --help            show this help message and exit
  -i IP                 ip address
  -p PORT               port
  -U USER               username
  -P PASSWORD           password
  -r ROUTE              REST route
  -X CMD                HTTP command (verb) default=GET
  -v, --verbose         increase output verbosity
/opt/RestApi/test #
```

or to get all data at once:

```
./testClient.py
```

or to get only the counting line data:

```
./testClient.py -r /apiv1/sensorData/counts
```

You could copy the testClient.py python script to your server and modify it to your needs. For this example it require python and curl to communicate with the device.

8.2 curl

To see how the REST API works the simple `curl` ⁺ command could be used. This `curl` as "client for URLs ⁺" is a command-line tool and library transferring data using various network protocols.

This `curl` command is included in most current operating systems. If you have no access to PC with `curl` in a command shell, you could also use the APS linux system - to do this, connect first to an APS via ssh ⁺.

The curl tool have a lot of options. In the following examples we use these:

curl options used in examples

- k = ignore certificates
- X = specify request method
- H = pass custom headers
- d = HTTP POST data
- o = write to file

To access data via the REST API, an authentication token must first be obtained from the sensor:

Example command to get authentication token

```
curl -k -X POST https://192.168.178.24:8091/auth -H "Content-Type: application/json" -d '{ "username": "admin", "password": "admin" }'
```

(Note: all examples in one line without line breaks)

The sensor responds with a token encapsulated in JSON⁺ (JavaScript Object Notation):

Example response with token

```
{ "access_token":  
"eyJhbGciOiJIUzUxMiIsInR5cCI6IkpXVCJ9.eyJpZGVudGl0eSI6MTAwMCwiaWF0IjoxNzAxNDM5MjY2LCJleHAiOiJlMDE0NDI4NjYsIm5iZiI6MTcwMTQzOTI2Nn0.mYlXIh6oxbxqYLfDPylRQT_MnkAVnxO8XtopRhOzRYKKRo9qgOs46Uqm5d4ICniltfNIaykFQp8_LeXVNwo8xw" }
```

This token is required and used in the extra header of following REST API data calls:

Example command to get all sensor data

```
curl -k -X GET https://192.168.178.24:8091/apiv2/sensorData -H "Authorization: Bearer  
eyJhbGciOiJIUzUxMiIsInR5cCI6IkpXVCJ9.eyJpZGVudGl0eSI6MTAwMCwiaWF0IjoxNzAxNDM5MjY2LCJleHAiOiJlMDE0NDI4NjYsIm5iZiI6MTcwMTQzOTI2Nn0.mYlXIh6oxbxqYLfDPylRQT_MnkAVnxO8XtopRhOzRYKKRo9qgOs46Uqm5d4ICniltfNIaykFQp8_LeXVNwo8xw"
```

(Note: all examples in one line without line breaks instead some white spaces)

The sensor responds with the data encapsulated in JSON (JavaScript Object Notation):

Example response with data

```
{
  "sensorData": {
    "macAddress": "00:0B:91:90:12:34",
    "hostname": "APS901234",
    "countingData": {
      "timeStamp": "2023-11-01T15:01:44+01:00",
      "countingLines": [
        {
          "name": "door",
          "data": [
            {
              "class": "Adult",
              "in": 84,
              "out": 83
            },
            {
              "class": "Child",
              "in": 1,
              "out": 1
            },
            {
              "class": "Group",
              "in": 85,
              "out": 84
            },
            {
              "class": "Cart",
              "in": 1,
              "out": 0
            },
            {
              "class": "Staff",
              "in": 0,
              "out": 0
            }
          ]
        },
        {
          "name": "gate",
          "data": [
            {
              "class": "Adult",
              "in": 8,
              "out": 7
            },
            {
              "class": "Child",
              "in": 1,
              "out": 0
            },
            {
              "class": "Group",
              "in": 9,
              "out": 7
            },
            {
              "class": "Cart",
              "in": 0,
              "out": 0
            },
            {
              "class": "Staff",
              "in": 0,
              "out": 0
            }
          ]
        }
      ]
    },
    "fillLevel": {
      "timeStamp": "2023-11-01T15:01:44+01:00",
      "zones": [
        {
          "name": "workshop",
          "data": [
            {
              "class": "Adult",
              "level": 0,
              "avgDwell": 0,
              "maxDwell": 0
            },
            {
              "class": "Child",
              "level": 0,
              "avgDwell": 0,
              "maxDwell": 0
            },
            {
              "class": "Cart",
              "level": 0,
              "avgDwell": 0,
              "maxDwell": 0
            },
            {
              "class": "Staff",
              "level": 0,
              "avgDwell": 0,
              "maxDwell": 0
            }
          ]
        }
      ]
    },
    "zoneStatistics": {
      "timeStamp": "2023-11-01T15:00:00+01:00",
      "interval": 300,
      "zones": [
        {
          "name": "workshop",
          "data": [
            {
              "class": "Adult",
              "count": 0,
              "maxFillLevel": 0,
              "maxDwell": 0,
              "meanDwell": 0
            },
            {
              "class": "Child",
              "count": 0,
              "maxFillLevel": 0,
              "maxDwell": 0,
              "meanDwell": 0
            },
            {
              "class": "Cart",
              "count": 0,
              "maxFillLevel": 0,
              "maxDwell": 0,
              "meanDwell": 0
            },
            {
              "class": "Staff",
              "count": 0,
              "maxFillLevel": 0,
              "maxDwell": 0,
              "meanDwell": 0
            }
          ]
        }
      ]
    },
    "zoneObjectList": {
      "timeStamp": "2023-11-01T15:00:00+01:00",
      "interval": 300,
      "zones": [
        {
          "name": "workshop",
          "objects": []
        }
      ]
    }
  }
}
```

Beside JSON data some REST API calls send files as e.g. image snapshots:

Example command to get image snapshots file

```
curl -k -X GET https://192.168.178.24:8091/apiv2/systemCommand/imageSnapshots -H
"Authorization: Bearer
eyJhbGciOiJIUzUxMiIsInR5cCI6IkpXVCJ9.eyJpZGVudGl0eSI6MTAwMCwiaWF0IjoxNzAxNDM5MjY2
LCJleHAiOiJlMDE0NDI4NjYsIm5iZiI6MTcwMTQzOTI2Nn0.mYlXIh6oxbxqYLfDPylRQT_MnkAVnx08X
topRhOzRYKKRo9qgOs46Uqm5d4ICniltfNIaykFQp8_LeXVNwo8xw" -o imageSnapshots.tar.gz
```

(Note: all examples in one line without line breaks instead some white spaces)

The received file need to be decompressed and the content files extracted from an archive by the following two commands:

Example to decompress and extract received files

```
gzip -d imageSnapshots.tar.gz
```

and

```
tar xf imageSnapshots.tar
```

8.3 Tool Insomnia

Insomnia is a free cross-platform desktop application that takes the pain out of interacting with HTTP-based APIs.

Download Insomnia from ➡ <https://insomnia.rest/download> and install it on your PC.

The standard setting of Insomnia is to check SSL certificates. The APS uses a self-generated certificate which cannot be validated. First disable the verification of SSL certificates.

1. ➡ Go to "Application → Preferences"(Fig. 2 /1).

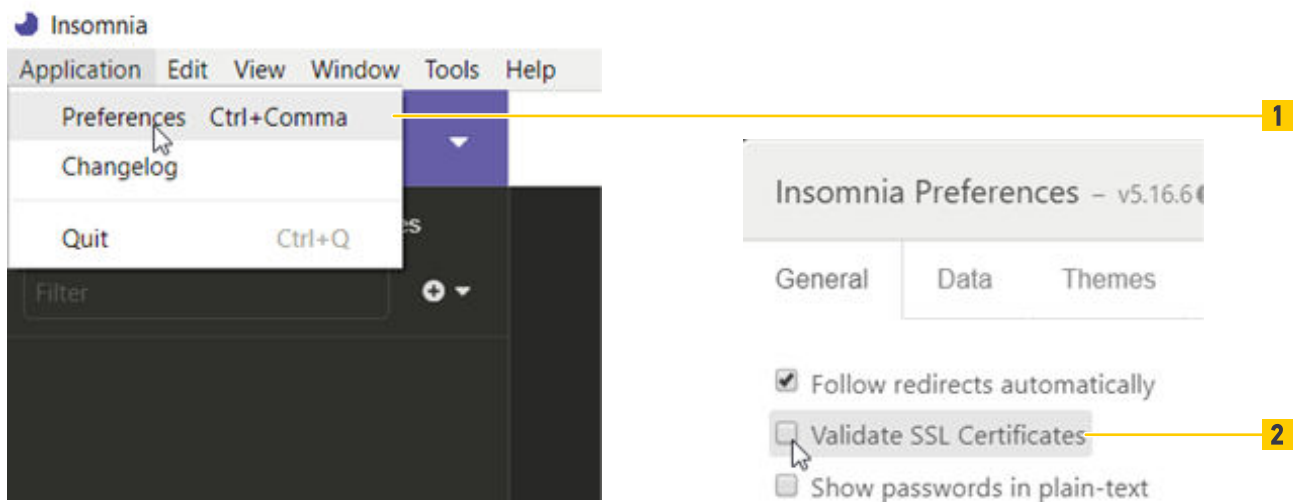


Fig. 2: Disable SSL certificates

2. ➡ Uncheck the box [Validate SSL Certificates] (Fig. 2 /2).

To get data from the APS an access token is needed for authentication.

1. ➡ To create a new request click on the plus symbol and select [New Request] (➡ Fig. 2 /1).

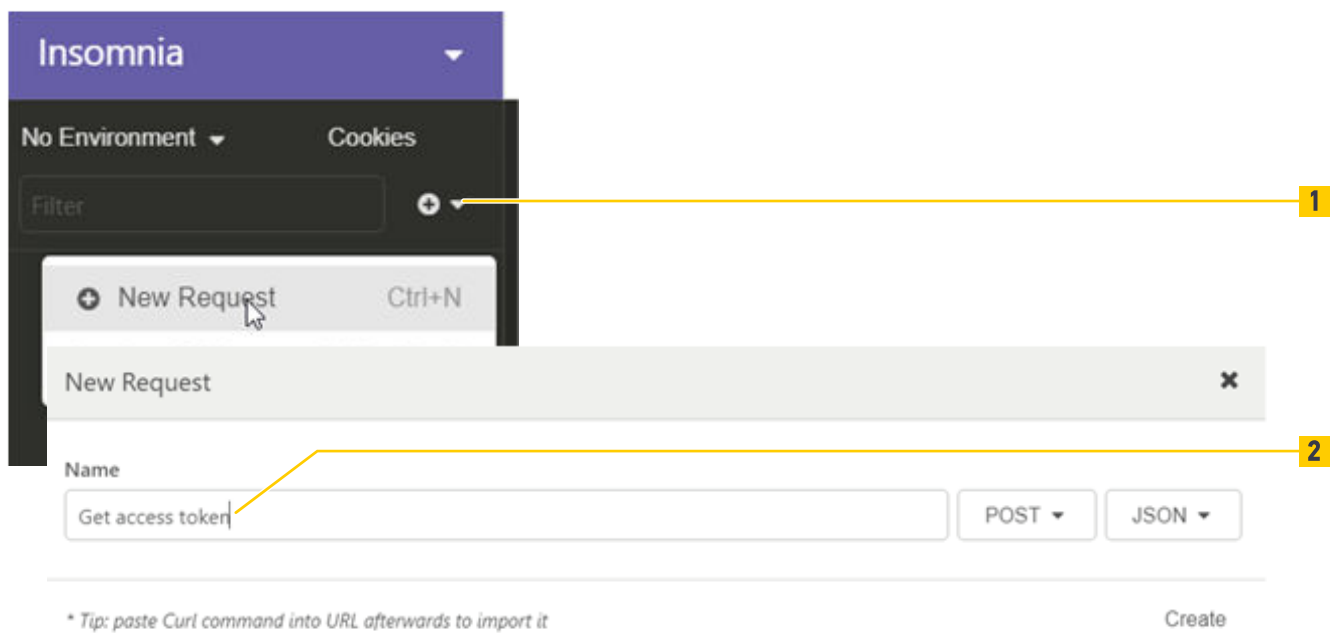


Fig. 3: Create new request

2. Name the request (→ Fig. 2 /2) and select [POST] and [JSON].
3. Add user name and password in the [JSON] (Fig. 4 /1).

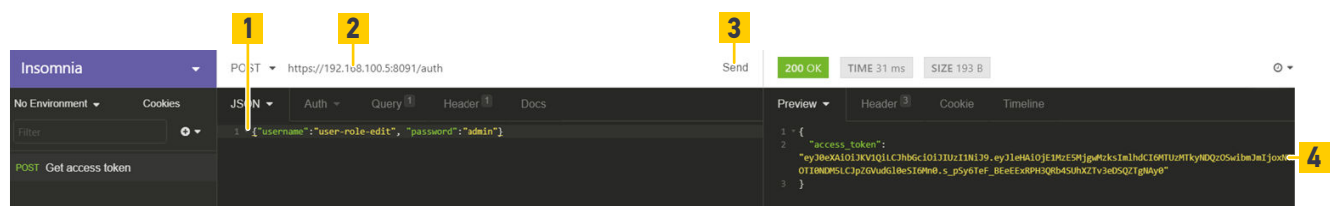


Fig. 4: Receive access token

4. Use the URL `https://[my-sensor-IP]:8091/auth` (Fig. 4 /2).
5. Press [Send] (Fig. 4 /3).
 - ➔ Receive the access token (Fig. 4 /4).

To get data from the APS use the received token.

1. Create a new request (→ Step 2 on page 78).

New Request

Name

Get all data

GET

* Tip: paste Curl command into URL afterwards to import it

Create

Fig. 5: Create new request GET

2. Name the request (Fig. 5 /1) and select [GET] (Fig. 5 /2).

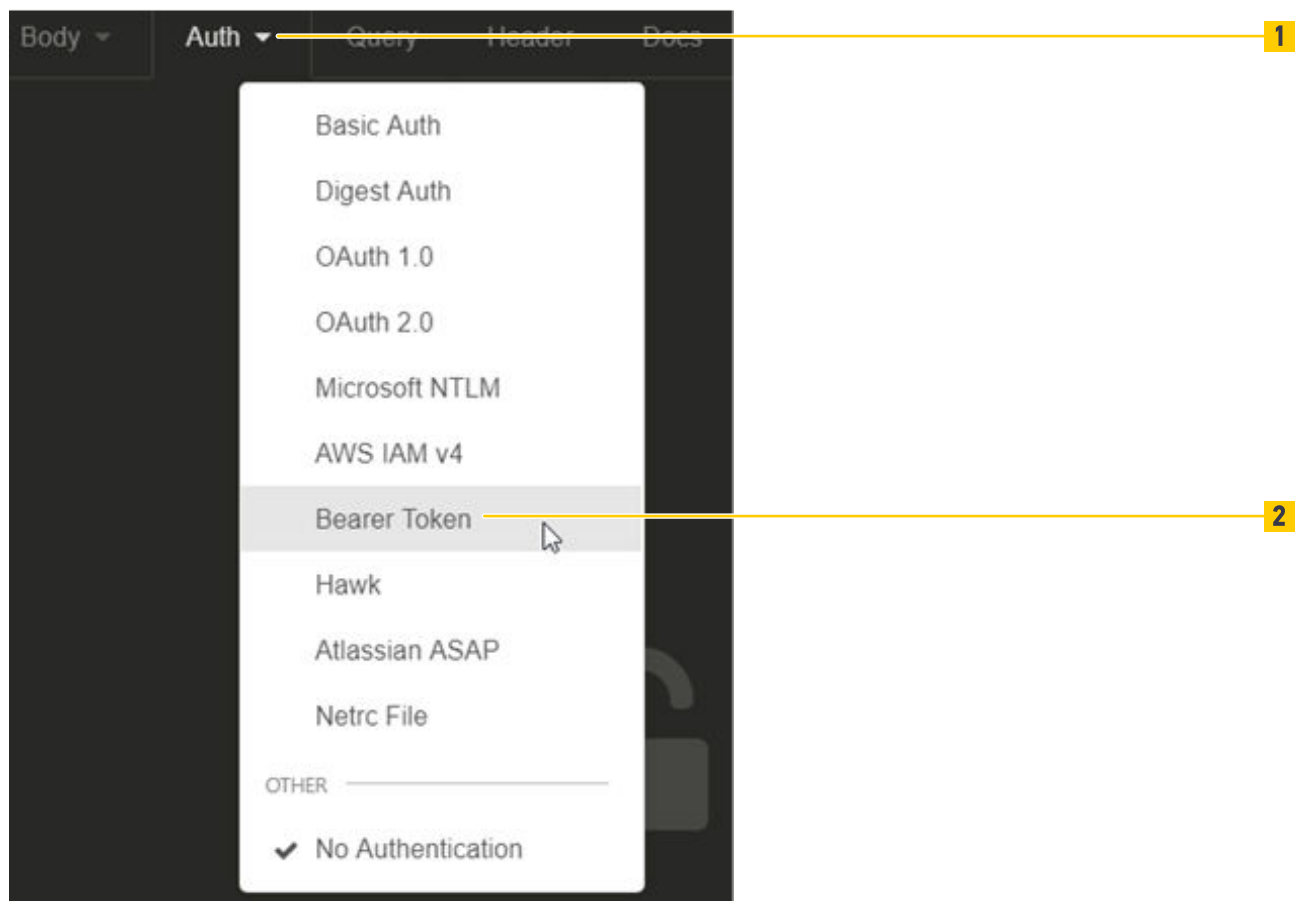


Fig. 6: Define authentication method

3. Go to the area [AUTH] (Fig. 6 /1) and select the method [Bearer Token] (Fig. 6 /2).
4. Copy the received token from the POST command.

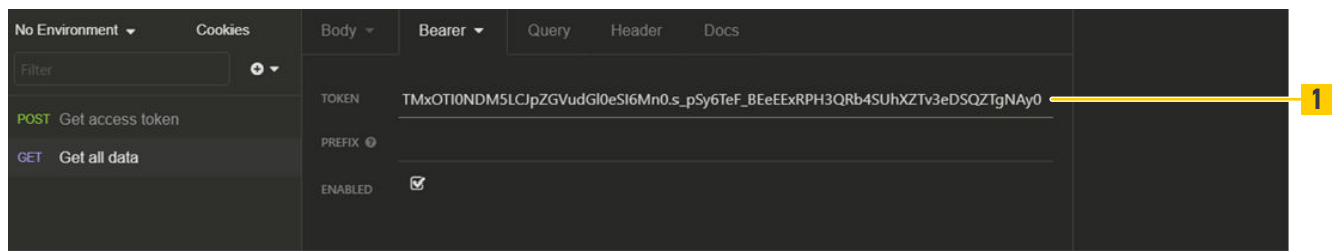


Fig. 7: Set the bearer token

5. Paste the token in the **[TOKEN]** field (Fig. 7 /1).
6. Send one of the described URL to select the kind of data
 - ➔ Receive the selected data.

Some examples to get data from the APS.

1. Send [https://\[my-sensor-IP\]:8091/apiv1/sensorData](https://[my-sensor-IP]:8091/apiv1/sensorData) (Fig. 8 /1).

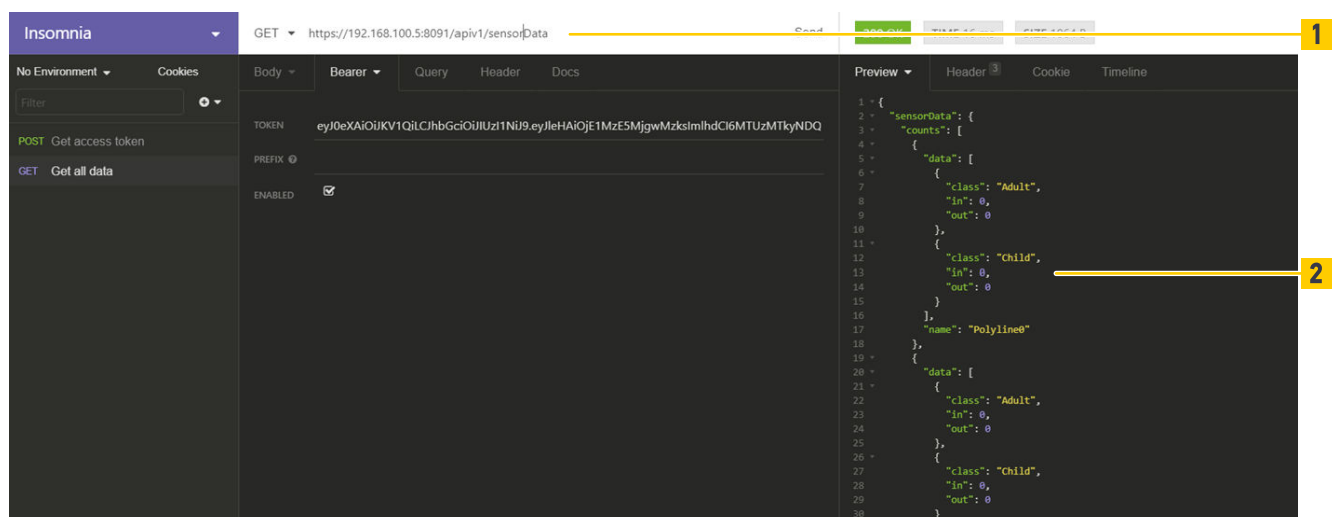


Fig. 8: GET all sensor data

- ➔ Receive all sensor data (Fig. 8 /2).

2. Send [https://\[my-sensor-IP\]:8091/apiv1/sensorData/counts](https://[my-sensor-IP]:8091/apiv1/sensorData/counts) (Fig. 9 /1).

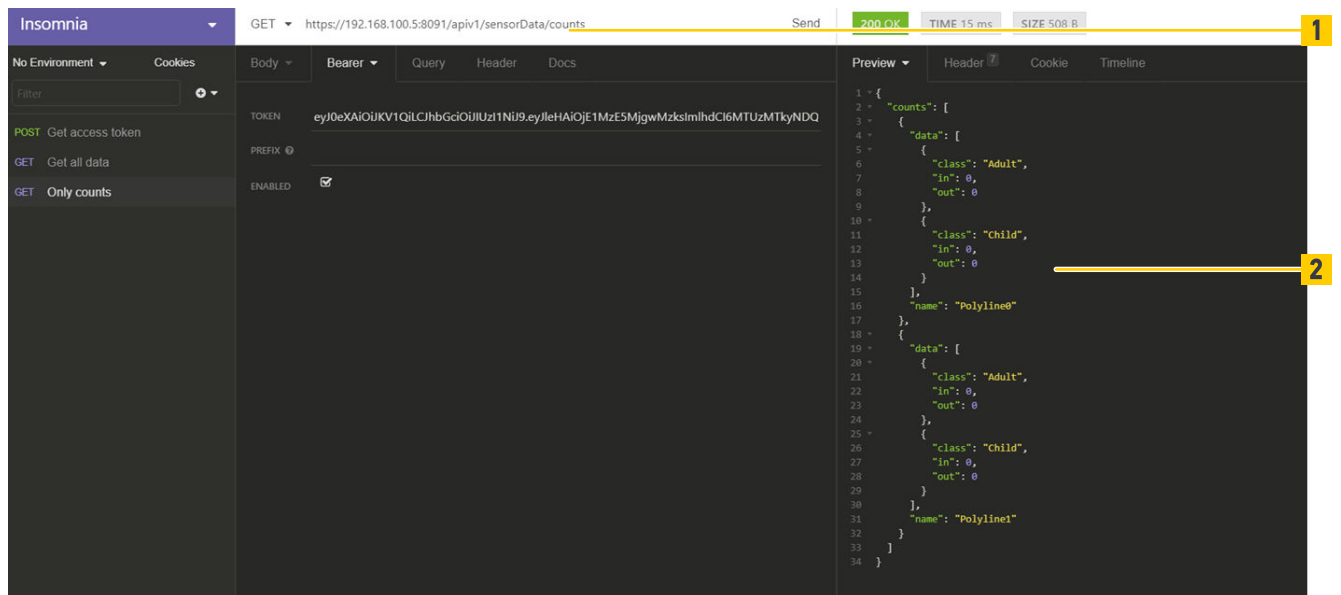


Fig. 9: GET counting data from all counting lines

➡ Receive all counting line data (Fig. 9 /2).

3. Send [https://\[my-sensor-IP\]:8091/apiv1/sensorData/counts/\[Name-of-my-line\]](https://[my-sensor-IP]:8091/apiv1/sensorData/counts/[Name-of-my-line]) (Fig. 10 /1).

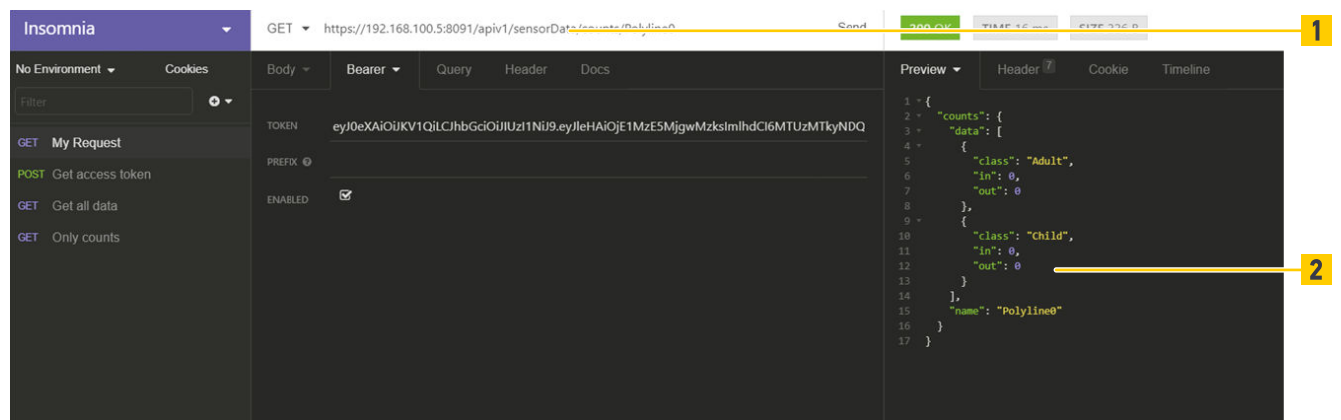


Fig. 10: GET counting data from selected counting line

➡ Receive counting data for single line (Fig. 10 /2).

9 APPENDIX

9.1 OSD Status Codes

The status of the Optical Self Diagnosis (OSD)⁺ `osd_state` is included in all messages that contain optically determined values and additional in start, alive and error messages. The `osd_state` contains one of the following values:

Status	Comments
OSD_STATE_OK	normal operation, no errors.
OSD_ERROR_COVERED	e.g. by a sticker, also only to one of the two camera sensors
OSD_ERROR_TOO_DARK	illumination is less than ~7 lux
OSD_ERROR_TOO_BRIGHT	more hypothetical, because direct sunlight and reflexions from sunlight do not cause "too bright"
OSD_ERROR	more hypothetical, general OSD error

9.2 HTTP Error Codes

Some responses are common HTTP communication states. They are used to confirm correct requests or provide feedback on unauthenticated or invalid requests:

Status Code	Description
200	OK
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not found
405	Method Not Allowed
429	Too Many Requests

10 GLOSSARY AND ABBREVIATIONS

APC	Automatic People Counter First generation people counter.
APS	Advanced People Sensor Second generation people sensor, successor of the APC ⁺ .
cURL	client for URL Command-line tool and library transferring data using various network protocols.
DHCP	Dynamic Host Configuration Protocol Protocol and service dynamically distributing network configuration parameters, such as IP address and servers.
HMI	Human Machine Interface e.g. the web-based configuration interface of the APS
HTTPS	Hypertext Transfer Protocol Secure Application protocol for secure communication in networks
JSON	JavaScript Object Notation Used by the APS REST interface.
MAC address	Media Access Control address Unique hardware address of a network device.
NAT	Network Address Translation Process of modifying IP address information during transit across a traffic routing device
OSD	Optical Self Diagnosis Software function for checking the visual range
SSH	The Secure SHell Protocol is a cryptographic network protocol for operating network services securely over an unsecured network.
URL	Uniform Resource Locator Reference to a web resource including protocol, server, file path, etc.
UWB	Ultra-Wideband Ultra-wideband is a radio technology that can use a very low energy level for short-range, high-bandwidth communications over a large portion of the radio spectrum.

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